

Problem

Determine the rms value for the waveform in Fig. 11.64.

Figure 11.64:



Step-by-step solution

Step 1 of 3

Refer to Figure 11.64 in the textbook.

Consider the following waveform:

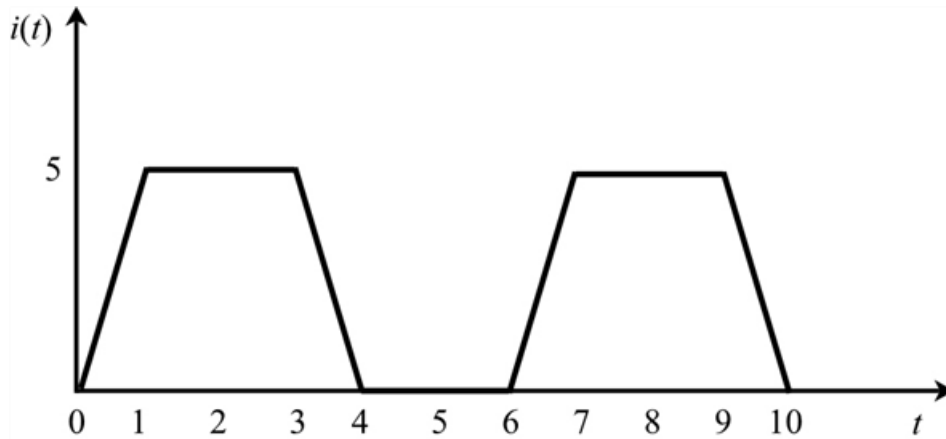


Figure 1

Step 2 of 3

The period of the Figure 1 is $T = 6$. Take the two points to calculate line for the negative slope. That is $(3, 5)$ and $(4, 0)$ for the interval of $3 < t < 4$ in Figure 1.

$$i - 5 = -5(t - 3)$$

$$i - 5 = -5t + 15$$

$$i = -5t + 20$$

Write the current waveform in Figure 1 in mathematical expression.

$$i(t) = \begin{cases} 5t, & 0 < t < 1 \\ 5, & 1 < t < 3 \\ -5t + 20, & 3 < t < 4 \\ 0, & 4 < t < 6 \end{cases} \dots (1)$$

Write the formula for rms value.

$$I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt} \dots (2)$$

Step 3 of 3

Substitute the equation (1) in equation (2) to calculate the rms value of the Figure 1.

$$\begin{aligned}
 I_{\text{rms}} &= \sqrt{\frac{1}{6} \left[\int_0^1 (5t)^2 dt + \int_1^3 (5)^2 dt + \int_3^4 (-5t + 20)^2 dt \right]} \\
 &= \sqrt{\frac{1}{6} \left[\int_0^1 25t^2 dt + \int_1^3 25 dt + \int_3^4 (25t^2 + 400 - 200t) dt \right]} \\
 &= \sqrt{\frac{1}{6} \left[25 \frac{t^3}{3} \Big|_0^1 + 25t \Big|_1^3 + \left(25 \frac{t^3}{3} + 400t - 200 \frac{t^2}{2} \right) \Big|_3^4 \right]} \\
 &= \sqrt{\frac{1}{6} \left[25 \frac{(1-0)}{3} + 25(3-1) + \left\{ \left[25 \left(\frac{64}{3} \right) + 400(4) - 200 \left(\frac{16}{2} \right) \right] - \left[25 \left(\frac{27}{3} \right) + 400(3) - 200 \left(\frac{9}{2} \right) \right] \right\} \right]}
 \end{aligned}$$

Simplify the expression further.

$$\begin{aligned}
 I_{\text{rms}} &= \sqrt{\frac{1}{6} \left[\frac{25}{3} + 50 + [533.33 + 1600 - 1600] - [225 + 1200 - 900] \right]} \\
 &= \sqrt{\frac{66.663}{6}} \\
 &= 3.333 \text{ A}
 \end{aligned}$$

Therefore, the rms value for the waveform in Figure 1 is 3.333 A.